Hindi Verbal Fluency: VFT and SpAM Analysis of Mental Lexicon Structure

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Abstract

This study examines Hindi verbal fluency and spatial similarity organization using a Verbal Fluency Task (VFT) and the Spatial Arrangement Method (SpAM). Thirty-five Hindi–English bilingual participants generated category members in four domains (animals, foods, colours, body-parts), yielding 712 valid Hindi/Hinglish responses (mean IRT 6490 ms, Skewness = 2.54). We evaluated semantic clustering, serial position effects, domain differences in productivity and IRT, SpAM alignment with semantic and phonetic embeddings, phonological similarity trends over retrieval order, and whether an integrated VFT+SpAM score outperforms VFT alone. Results show a robust semantic boundary cost (between-cluster IRT > within-cluster IRT; t = 9.30, d = 1.12, p ¡ 0.001) and modest semantic alignment in SpAM (mean ARI = 0.1765), while evidence for phonological effects and domain-level differences is weak. The integrated score does not outperform the VFT-only score. Overall, Hindi lexical retrieval appears structured by semantic neighborhoods with limited phonological facilitation.

1 Introduction

The Verbal Fluency Task (VFT) requires participants to generate category members (e.g., "Animals") within 60 seconds. Responses tend to cluster by semantic subcategory, with pauses when switching between clusters, consistent with the clustering-and-switching model [4]. Hindi adds three complications to standard VFT analysis: (1) Hindi typing demands inflate IRTs, (2) educated participants code-switch into English mid-trial, and (3) culturally grounded domains such as foods and animals may alter cluster structure. This study examines how Hindi speakers search their mental lexicon using VFT and SpAM, and whether spatial similarity maps align with semantic and phonetic models.

2 Related Work

Phonological similarity influences semantic memory retrieval, suggesting that sound structure can bias access to meaning [1]. Words that are phonologically similar tend to be semantically related more than expected by chance. The Spatial Arrangement Method efficiently captures human semantic similarity judgments by arranging items based on perceived relatedness [2].

3 Hypotheses

H1: Between-cluster IRT is greater than within-cluster IRT (semantic boundary cost).
H2: IRT increases with serial position (lexical exhaustion).
H3: Total word count differs across domains.
H4: Mean IRT differs across domains.
H5: SpAM clusters align with semantic embeddings above chance.
H6: SpAM clusters align with phonetic embeddings above chance.
H7: Phonological-over-semantic similarity increases over retrieval order.
H8: Higher phonological similarity predicts higher productivity.
H9: Integrated VFT+SpAM score correlates differently than VFT-only.

4 Research Questions (RQ1-RQ5)

RQ1: Do semantic categories differ in productivity (words) or retrieval speed (IRT)? (H1, H2, H3, H4)
RQ2: Does participant spatial structure align with semantic or phonetic models? (H5, H6)
RQ3: Does an integrated VFT+SpAM score outperform VFT-only? (H9)
RQ4: Does phonological-over-semantic similarity increase over retrieval order? (H7)
RQ5: Does phonological similarity predict lexical productivity? (H8)

5 Methodology

5.1 Participants and Design

Within-subjects repeated-measures design; each of 35 participants completed all four semantic domains. Variable summary: N = 35; Gender: 32 Male, 3 Female; Age: M = 23.1 yrs, SD = 1.9, range 19–27; Education: M = 16.5 yrs, SD = 1.7, range 14–20; States (India): 14 states; Gujarat (7), MP (6), Bihar (5), ...

5.2 Stimuli, Apparatus, and Procedure

Four domains were chosen to span vocabulary size and structure: Animals (open, large), Foods (open, culturally rich), Colours (closed, 15 items), Body-parts (semi-closed). Responses were collected via a custom web app logging cumulative timestamps to the nearest 100 ms. Participants typed category members in 60 s per domain (Hindi cue, e.g., "Jaanwar"), preceded by a 1-minute Furniture warm-up. Domain order was fixed: Animals - Foods - Colours - Body-parts.

5.3 Processing Pipeline

Raw responses.json was processed in five steps: (1) IRT computation ($\text{IRT}_i = t_i - t_{i-1}$), where t_i is the cumulative timestamp of the i-th response. (2) language tagging. (3) Hindi filter retaining 712 of 1340 tokens (53%). (4) adaptive cluster scoring - switch when IRT > (mean + 1 SD) per participant. (5) export to `merged_vft_spam_responses_enriched.csv`.

5.4 Dataset Variables

Column	Description
subject_id	Anonymised participant identifier
process_id	SpAM session id (where available)
domain	Animals / Foods / Colours / Body-parts
word	Word as typed (original script)
rt_ms	Inter-response time (ms)
position	Serial position within trial (1 = first word)
language_type	Hindi/Hinglish or English

Derived per participant: within-cluster IRT (mean IRT where is_switch = False), between-cluster IRT (mean IRT where is_switch = True), mean cluster size (mean words per cluster), and total fluency score (count of Hindi tokens).

5.5 Models and Statistical Tests

Semantic patterns were modeled using the multilingual sentence-transformer `google/muril-base-cased` on original words [3]. Phonetic patterns were modeled by transliterating words into phonetic keys and embedding them in the same space. SpAM cluster layouts were compared with semantic and phonetic cluster structures using ARI and NMI.

We used Welch's t-test (H1), linear mixed-effects models per domain (H2: IRT $\sim$ position + (1 + position | subject)), Kruskal–Wallis tests (H3, H4), binomial sign tests and permutation tests for ARI (H5, H6), a mixed-effects model for similarity type by position (H7: similarity

$\sim$ type * position + (1 + position | subject)), Spearman correlation (H8), and Steiger's Z for dependent correlations (H9).

6 Hypothesis Test Summary (H1–H9)

Table 1: Consolidated hypothesis outcome summary across all nine hypotheses.

Hyp.	Prediction	Key Result	Signif.	Decision
H1	Between > within IRT	$t = 9.30$, $d = 1.12$	$p < 0.001$	**Supported**
H2	Positive serial slowing	No positive slope (0/4)	$p > 0.05$	Not supported
H3	Domain effect on words	$H = 3.264$	$p = .196$	Not supported
H4	Domain effect on IRT	$H = 3.903$	$p = .142$	Not supported
H5	Semantic ARI > 0	68/86 pos.; ARI = 0.177	sign<.001	Mixed
H6	Phonetic ARI > 0	52/86 pos.; ARI = 0.075	sign = .066	Not supported
H7	Phon-over-sem increases	$\beta = +0.003$	$p = .091$	Not supported
H8	Phon. sim. → WC	$\rho = +0.114$	$p = .257$	Not supported
H9	Integrated vs VFT-only	Steiger $z = 0.000$	$p = 1.000$	Not supported

7 Results

7.1 H1: Semantic Clustering (Between vs Within IRT)

Between-cluster pauses are higher than within-cluster pauses, supporting structured retrieval and a strong semantic boundary cost. The effect size is large (d = 1.12), indicating that semantic boundaries produce meaningful slowdowns rather than marginal timing differences.

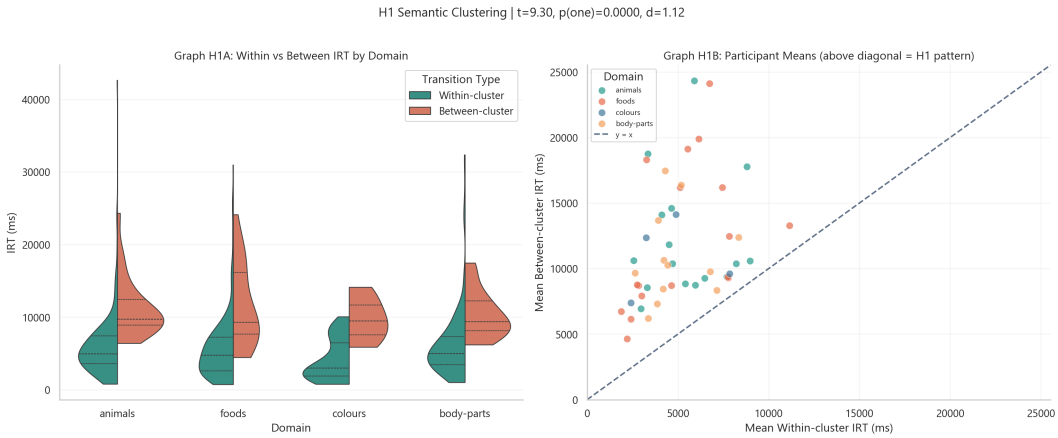

Figure 1: Between-cluster IRT > within-cluster IRT (plot).

7.2 H2: Lexical Exhaustion (Serial Position)

No positive slope was detected across domains. Linear mixed-effects models do not support a consistent serial slowing effect, suggesting that participants do not uniformly slow down as retrieval progresses.

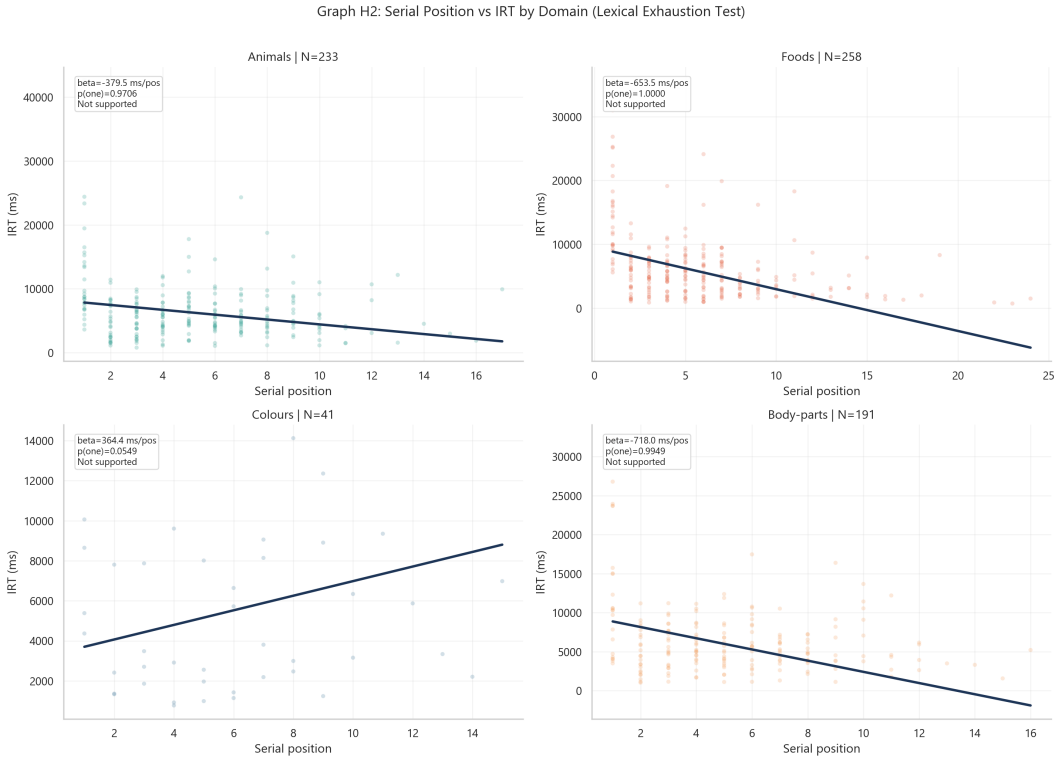

Figure 2: Serial position effect by domain (LME).

7.3 H3 and H4: Domain Differences

Domain-level differences in word output and retrieval time are not statistically significant in this sample, indicating that domain choice does not reliably alter productivity or mean IRT under the current data constraints.

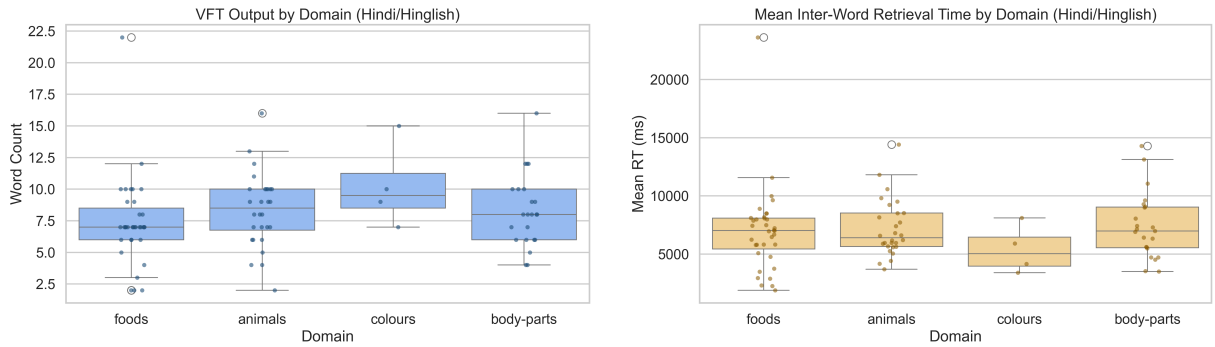

Figure 3: Domain differences in productivity (left) and retrieval speed (right).

7.4 H5 and H6: SpAM Alignment With Semantic and Phonetic Structure

Semantic ARI tends to be above phonetic ARI, but the effect is modest. The sign test suggests a positive semantic signal, while permutation results indicate that the effect is not robust against chance alignment. Phonetic alignment remains weak and statistically unsupported.

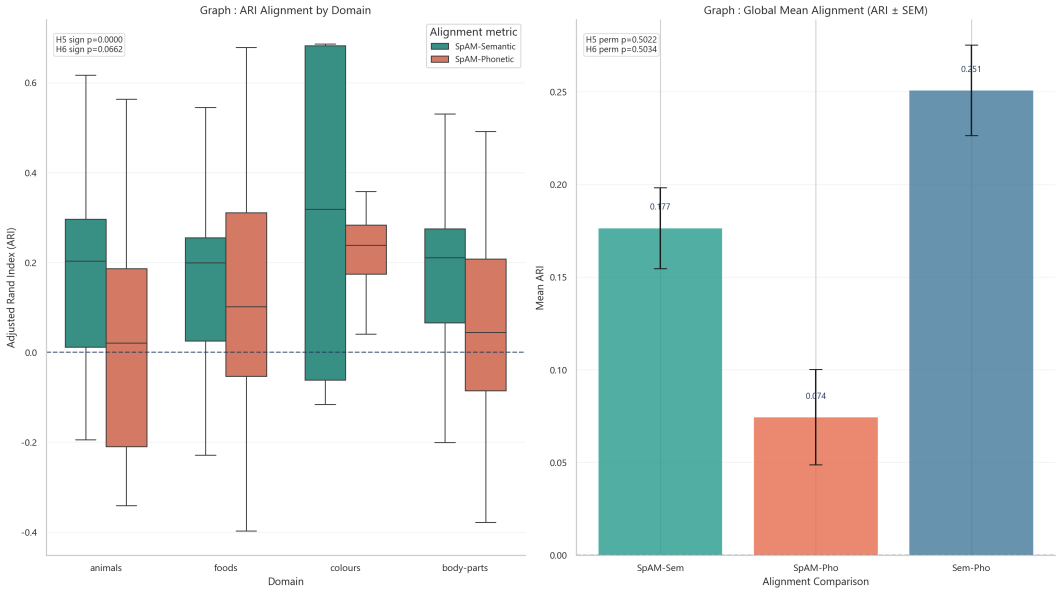

Figure 4: Cluster alignment: semantic vs phonetic ARI.

7.5 H7: Semantic vs Phonological Similarity Over Retrieval Order

No reliable phonological facilitation trend is observed over retrieval order. The interaction term is positive but non-significant (beta = 0.003127, p = 0.0905), indicating only weak evidence for increasing phonological dominance.

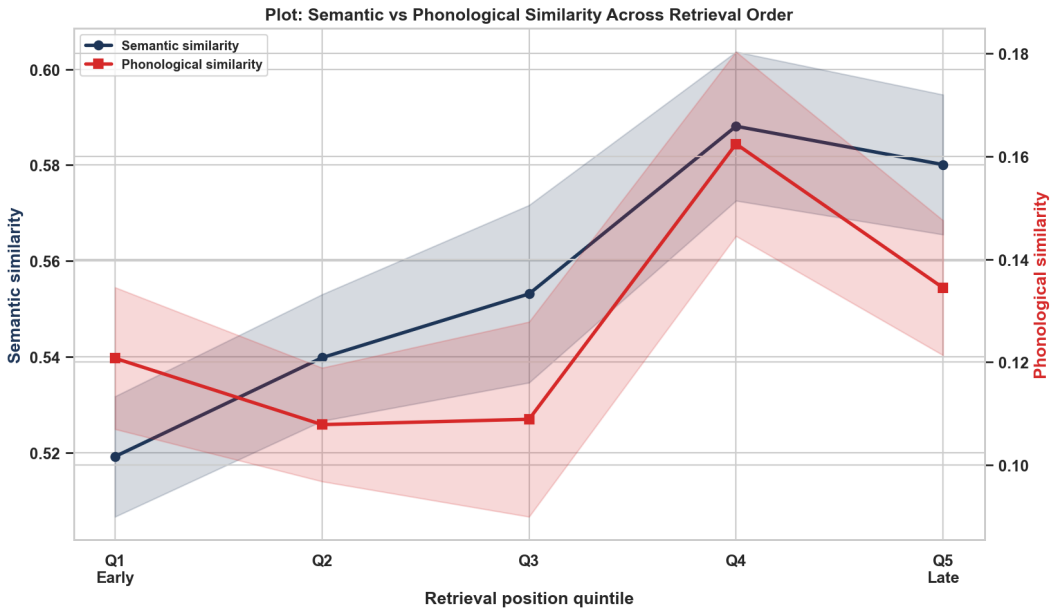

Figure 5: Semantic vs phonological similarity across retrieval order.

7.6 H8: Phonological Similarity and Productivity

Higher phonological similarity did not predict higher productivity (Spearman rho = 0.114, p(one) = 0.257). The effect is small and inconsistent with a strong facilitation account.

Interpretation: Semantic neighbourhood density does not predict retrieval latency. IRT is governed by *active clustering strategy* (H1) rather than passive neighbourhood geometry. This reinforces that bilingual retrieval dynamics differ from monolingual phonological neighbourhood effects [1].

7.7 H9: Integrated VFT + SpAM vs VFT-Only

The combined VFT+SpAM score does not outperform the VFT-only score; the two correlations are identical (Steiger z = 0.000). This indicates that SpAM contributes limited incremental predictive value for confidence in this dataset.

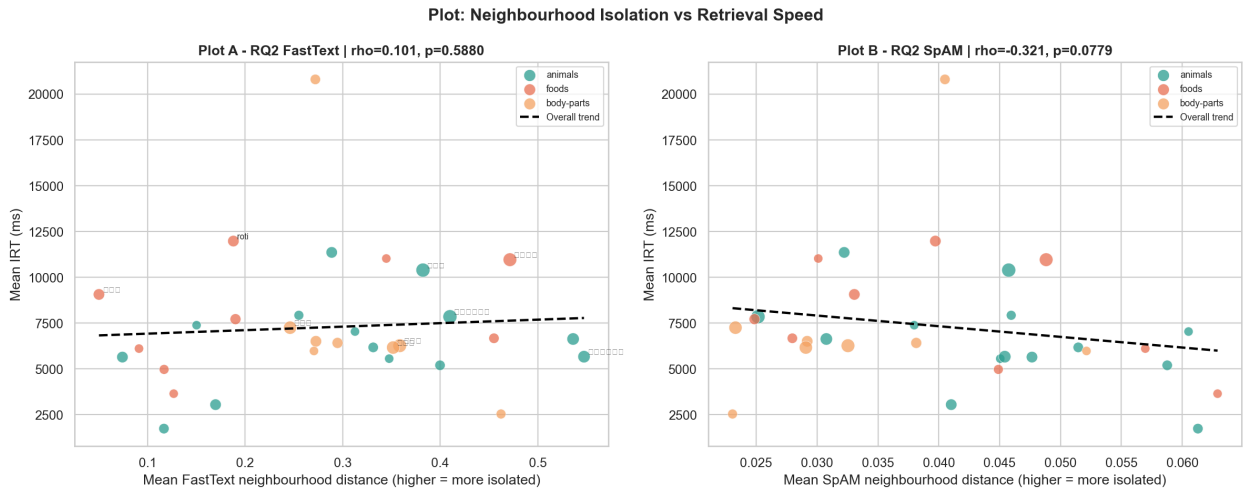

Figure 6: Phonological neighbourhood density vs IRT.

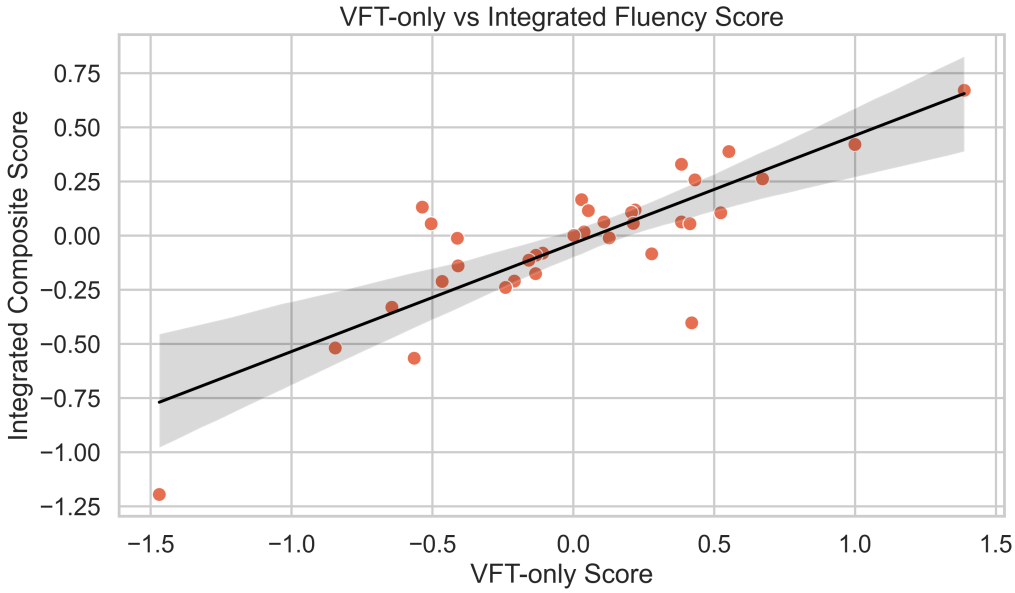

Figure 7: Integrated score vs VFT-only correlation with confidence.

8 Discussion

The strongest evidence supports semantic clustering: within-cluster IRTs are shorter than between-cluster IRTs, indicating a clear boundary cost in Hindi retrieval. SpAM layouts align weakly with semantic embeddings, suggesting partial semantic structure in spatial similarity judgments. In contrast, phonological effects are weak or inconsistent across tests, and domain-level differences in productivity and IRT do not reach statistical significance. The integrated VFT+SpAM score does not improve over VFT-only, indicating that SpAM

contributes limited incremental predictive value in this sample.

9 Limitations

The dataset is small and includes substantial English code-switching, which reduces the effective Hindi sample. The colours domain is underrepresented, limiting statistical power and generalizability for domain comparisons.

10 Conclusion

Hindi lexical retrieval is organized primarily by semantic neighborhoods rather than random retrieval. SpAM captures modest semantic structure, while phonological organization is not strongly supported. These findings are consistent with clustering-and-switching models and highlight semantic boundaries as the dominant constraint in Hindi verbal fluency.

11 Contributions

Hypothesis-based split: H1, H4, H7 (Akshat) — H2, H5, H8 (Om) — H3, H6, H9 (Ankit). Each member contributed to analysis and writing using shared frameworks (permutation tests, clustering) with weekly integration checkpoints.

Code and Data Availability

All analysis code, raw data, and figures are openly available in the project GitHub repository: `https://github.com/omii2403/Hindi-Fluency`. The repository contains: responses.json (raw session data), `merged_vft_spam_responses_enriched.csv` (processed dataset), `Hindi_fluency_final` (all analyses and figures), and this report source (Report_Final.tex).

References

[1] Isabelle Dautriche, Kyle Mahowald, Edward Gibson, Anne Christophe, and Steven T. Piantadosi. Wordform similarity increases with semantic similarity. *Cognitive Science*, 41(8):2149–2169, 2016.

[2] Michael C. Hout, Megan H. Papesh, and Stephen D. Goldinger. The versatility of SpAM. *Journal of Experimental Psychology: General*, 142(1):256–281, 2013. doi: 10.1037/a0028865.

[3] Abhilasha A. Kumar et al. Verbal fluency in bilingual South Asian older adults. *Journal of the International Neuropsychological Society*, 28(4):412–421, 2022. doi: 10.1017/S135561772100057X.

[4] Angela K. Troyer, Morris Moscovitch, and Gordon Winocur. Clustering and switching as two components of verbal fluency. *Neuropsychology*, 11(1):138–146, 1997. doi: 10.1037/0894-4105.11.1.138.